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Subject: Mathematics **WEEKLY CHAPTER TEST – 01 (SOLUTIONS)** **Time:** 45 Minutes
Class: Class X (CBSE) **Chapter 6: Triangles** **MM:** 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit theorem/formula declaration, neat geometric diagrams, logical reasoning, and final boxed results.
2. In descriptive proofs, state **Given**, **To Prove**, **Construction** (where needed), and justify every equality or similarity criterion.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M]**. Alternative valid methods receive full credit.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice Questions

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Q1. Two polygons of the same number of sides are similar if: [1]

(a) Their corresponding angles are equal and corresponding sides are equal
(b) Their corresponding angles are proportional and sides are equal
(c) Their corresponding angles are equal and corresponding sides are proportional
(d) Their perimeters and areas are equal

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) Their corresponding angles are equal and corresponding sides are proportional

We know the fundamental definition of similarity of polygons:

Formula / Theorem Used: Two polygons having the same number of sides are similar if (i) their corresponding angles are equal, and (ii) their corresponding sides are in the same ratio (proportional).

Option (a) defines congruence (sides equal, not merely proportional), option (b) reverses the conditions incorrectly, and option (d) alone does not guarantee identical shape. Hence, option **(c)** is correct.

Q2. Which of the following geometric figures are **always** similar to each other? [1]

(a) Two rectangles (b) Two rhombuses (c) Two equilateral triangles (d) Two isosceles triangles

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) Two equilateral triangles

We know the AAA similarity condition for triangles:

Formula / Theorem Used: If corresponding angles of two triangles are equal, their corresponding sides are proportional and the triangles are similar.

In any equilateral triangle, all three interior angles are always 60° and all three sides are equal (so the ratio of corresponding sides between any two equilateral triangles is constant). Rectangles can have different length-to-breadth ratios, rhombuses can have different angles, and isosceles triangles can have different vertex angles. Hence, two equilateral triangles are **always** similar.

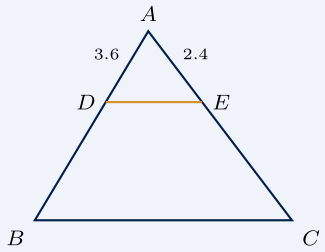
Q3. In $\triangle ABC$, D and E are points on sides AB and AC respectively such that $DE \parallel BC$. If $AD = 3.6$ cm, $AB = 9.6$ cm, and $AE = 2.4$ cm, then the length of AC is: [1]

(a) 6.4 cm (b) 5.4 cm (c) 4.0 cm (d) 7.2 cm

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (a) 6.4 cm

Given: In $\triangle ABC$, $DE \parallel BC$, $AD = 3.6$ cm, $AB = 9.6$ cm, and $AE = 2.4$ cm.



We know the corollary of the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: If $DE \parallel BC$ in $\triangle ABC$, then $\dfrac{AD}{AB} = \dfrac{AE}{AC}$.

Substituting the given lengths into the proportion:

$$\frac{3.6}{9.6} = \frac{2.4}{AC}$$

Simplifying $\dfrac{3.6}{9.6} = \dfrac{36}{96} = \dfrac{3}{8}$ (dividing numerator and denominator by 12):

$$\frac{3}{8} = \frac{2.4}{AC} \implies 3 \times AC = 2.4 \times 8 = 19.2$$

Dividing both sides by 3:

$$AC = \frac{19.2}{3} = 6.4 \text{ cm} \implies \boxed{AC = \mathbf{6.4} \text{ cm}}$$

Q4. In $\triangle ABC$, $DE \parallel BC$ such that $AD = x$, $DB = x - 2$, $AE = x + 2$, and $EC = x - 1$. The value of x is: [1]

(a) 3 (b) 4 (c) 5 (d) 6

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: (b) 4

Given: In $\triangle ABC$, $DE \parallel BC$, $AD = x$, $DB = x - 2$, $AE = x + 2$, and $EC = x - 1$.

We know the Basic Proportionality Theorem (Thales' Theorem):

Formula / Theorem Used: If a line is drawn parallel to one side of a triangle intersecting the other two sides, then $\dfrac{AD}{DB} = \dfrac{AE}{EC}$.

Substituting the given algebraic expressions:

$$\frac{x}{x-2} = \frac{x+2}{x-1}$$

Cross-multiplying both sides:

$$x(x-1) = (x+2)(x-2)$$

Expanding both sides using $(a+b)(a-b) = a^2 - b^2$ on the right-hand side:

$$x^2 - x = x^2 - 4$$

Subtracting x^2 from both sides:

$$-x = -4 \implies x = 4$$

Checking positivity of side segments for $x = 4$: $AD = 4 > 0$, $DB = 2 > 0$, $AE = 6 > 0$, $EC = 3 > 0$. Thus, $\boxed{x = 4}$.

Q5. **Assertion (A):** Two congruent figures are always similar.
Reason (R): Similar figures always have the same size and the same shape. [1]

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (c) Assertion (A) is true but Reason (R) is false.

We know the distinction between congruence and similarity of geometric figures:

Formula / Theorem Used: Congruent: Same shape and same size (1 : 1 side ratio). **Similar:** Same shape (size may differ).

Evaluating (A): Every pair of congruent figures has equal corresponding angles and corresponding sides in the exact ratio 1 : 1. Therefore, congruent figures are always similar. Hence, **Assertion (A) is true**.

Evaluating (R): Similar figures must have the same shape, but they do *not* necessarily have the same size (one can be a scaled enlargement or reduction of the other). Hence, **Reason (R) is false**.

SECTION B: Descriptive Questions MATHS MANTHAN | (1 × 2 + 3 × 3 = 11 Marks)

Q6. In $\triangle ABC$, D and E are points on sides AB and AC respectively. If $AD = 5.7$ cm, $BD = 9.5$ cm, $AE = 3.3$ cm, and $AC = 8.8$ cm, determine whether $DE \parallel BC$, giving mathematical justification. [2]

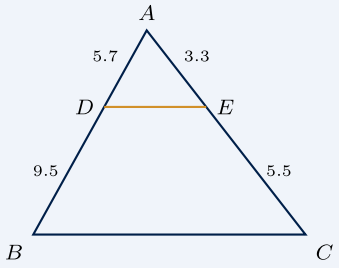
► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Finding Segment EC & Theorem [0.5 Mark]

Given: In $\triangle ABC$, $AD = 5.7$ cm, $BD = 9.5$ cm, $AE = 3.3$ cm, and $AC = 8.8$ cm.

Since E lies on side AC:

$$EC = AC - AE = 8.8 - 3.3 = 5.5 \text{ cm}$$



We know the Converse of Basic Proportionality Theorem:

Formula / Theorem Used: If a line divides any two sides of a triangle in the same ratio, i.e., $\frac{AD}{DB} = \frac{AE}{EC}$, then the line is parallel to the third side ($DE \parallel BC$).

Step 2: Computing Both Ratios Step by Step [1 Mark]

Computing the first ratio on side AB:

$$\frac{AD}{DB} = \frac{5.7}{9.5} = \frac{57}{95} = \frac{3 \times 19}{5 \times 19} = \frac{3}{5}$$

Computing the second ratio on side AC:

$$\frac{AE}{EC} = \frac{3.3}{5.5} = \frac{33}{55} = \frac{3 \times 11}{5 \times 11} = \frac{3}{5}$$

Step 3: Comparison & Conclusion [0.5 Mark]

Since $\frac{AD}{DB} = \frac{AE}{EC} = \frac{3}{5}$, the line DE divides sides AB and AC in the exact same ratio.

$\therefore$ **DE $\parallel$ BC** (by Converse of Basic Proportionality Theorem)

Q7. State Basic Proportionality Theorem (Thales' Theorem). Using this theorem, prove that a line drawn through the midpoint of one side of a triangle parallel to another side bisects the third side. **[3]**

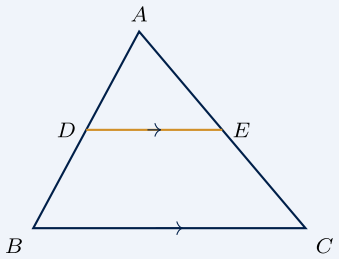
▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Statement of Basic Proportionality Theorem (Thales' Theorem) [1 Mark]

We state the Basic Proportionality Theorem:

Formula / Theorem Used: If a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points, then the other two sides are divided in the same ratio.

Step 2: Formal Setup (Given & To Prove) [0.5 Mark]



Given: In $\triangle ABC$, D is the midpoint of side AB (so $AD = DB$), and a line DE drawn through D parallel to BC ($DE \parallel BC$) intersects side AC at point E.

To Prove: Line DE bisects side AC, i.e., $AE = EC$ (or E is the midpoint of AC).

Step 3: Deductive Proof Using BPT [1.5 Marks]

1. Since D is the midpoint of AB, we have:

$$AD = DB \implies \frac{AD}{DB} = 1 \qquad \text{— (i)}$$

2. In $\triangle ABC$, it is given that $DE \parallel BC$. Applying the Basic Proportionality Theorem:

$$\frac{AD}{DB} = \frac{AE}{EC} \qquad \text{— (ii)}$$

3. Substituting $\frac{AD}{DB} = 1$ from equation (i) into equation (ii):

$$1 = \frac{AE}{EC}$$

Cross-multiplying both sides by EC:

$$AE = EC$$

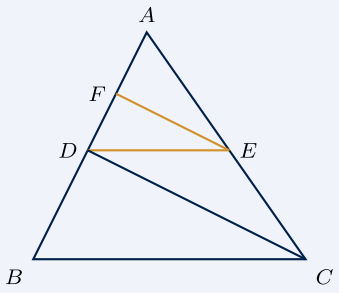
Therefore, point E is the midpoint of side AC, meaning line DE bisects the third side AC.

$$\boxed{\therefore AE = EC \quad \text{[Hence Proved]}}$$

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Q8. In the given figure, $DE \parallel BC$ and $CD \parallel EF$. Prove mathematically that $AD^2 = AB \times AF$. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: In $\triangle ABC$, D and F lie on AB, E lies on AC, such that $DE \parallel BC$ and $CD \parallel EF$ (i.e., $EF \parallel DC$).
To Prove: $AD^2 = AB \times AF$.

We know the Basic Proportionality Theorem (Thales' Theorem) and its whole-side corollary:

Formula / Theorem Used: If a line is drawn parallel to one side of a triangle, it divides the other two sides in the same ratio.

Step 1: Applying BPT in $\triangle ABC$ ($DE \parallel BC$) [1 Mark]
In $\triangle ABC$, since $DE \parallel BC$, by BPT:

$$\frac{DB}{AD} = \frac{EC}{AE}$$

Adding 1 to both sides to form the full sides AB and AC:

$$\frac{DB}{AD} + 1 = \frac{EC}{AE} + 1 \implies \frac{DB + AD}{AD} = \frac{EC + AE}{AE}$$

Since $AD + DB = AB$ and $AE + EC = AC$, taking reciprocals gives:

$$\frac{AB}{AD} = \frac{AC}{AE} \implies \frac{AD}{AB} = \frac{AE}{AC} \quad \text{— (i)}$$

Step 2: Applying BPT in $\triangle ADC$ ($EF \parallel DC$) [1 Mark]
In $\triangle ADC$, since $EF \parallel DC$ (given $CD \parallel EF$), by BPT:

$$\frac{FD}{AF} = \frac{EC}{AE}$$

Adding 1 to both sides:

$$\frac{FD}{AF}+1 = \frac{EC}{AE}+1 \implies \frac{FD + AF}{AF} = \frac{EC + AE}{AE} \implies \frac{AD}{AF} = \frac{AC}{AE}$$

Taking reciprocals of both sides:

$$\frac{AF}{AD} = \frac{AE}{AC} \quad \text{— (ii)}$$

Step 3: Equating Relations & Cross-Multiplying [1 Mark]
From equations (i) and (ii), both left-hand sides equal $\frac{AE}{AC}$, so:

$$\frac{AD}{AB} = \frac{AF}{AD}$$

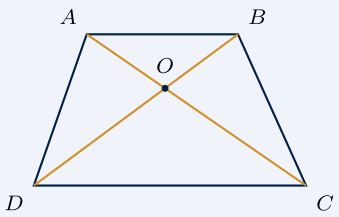
Cross-multiplying:

$$AD \times AD = AB \times AF \implies \boxed{\therefore AD^2 = AB \times AF \quad \text{[Hence Proved]}}$$

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Q9. In a trapezium $ABCD$ with $AB \parallel DC$, diagonals AC and BD intersect each other at the point O. Using the concept of similarity of triangles or Basic Proportionality Theorem, show that $\dfrac{AO}{OC} = \dfrac{BO}{OD}$. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Given: A trapezium $ABCD$ in which $AB \parallel DC$. Diagonals AC and BD intersect each other at point O.
To Prove: $\dfrac{AO}{OC} = \dfrac{BO}{OD}$.
We know the Angle-Angle (AA) criterion for similarity of triangles:

> **Formula / Theorem Used:** If two angles of one triangle are respectively equal to two angles of another triangle, then the two triangles are similar, and their corresponding sides are proportional.

Step 1: Equal Angles in $\triangle AOB$ and $\triangle COD$ [1.5M]
In $\triangle AOB$ and $\triangle COD$:

1. $\angle OAB = \angle OCD$ (Alternate interior angles, since $AB \parallel DC$ with transversal AC)
2. $\angle OBA = \angle ODC$ (Alternate interior angles, since $AB \parallel DC$ with transversal BD)
3. $\angle AOB = \angle COD$ (Vertically opposite angles)

Step 2: Similarity & Side Proportion [1.5M]
By the **AA (or AAA) similarity criterion**, we obtain:

$$\triangle AOB \sim \triangle COD$$

Since corresponding sides of similar triangles are proportional:

$$\frac{AO}{CO} = \frac{BO}{DO} \implies \boxed{\therefore \frac{AO}{OC} = \frac{BO}{OD} \quad \text{[Hence Proved]}}$$

▶ ALTERNATIVE METHOD (ALITER — USING BASIC PROPORTIONALITY THEOREM):
Construction: Through O, draw a line $OE \parallel AB$ meeting AD at E. Since $AB \parallel DC$, we also have $OE \parallel DC$.
1. In $\triangle ADC$, since $EO \parallel DC$, by BPT: $\dfrac{AE}{ED} = \dfrac{AO}{OC}$ — (i).
2. In $\triangle DAB$, since $EO \parallel AB$, by BPT: $\dfrac{DE}{EA} = \dfrac{DO}{OB}$, and taking reciprocals gives $\dfrac{AE}{ED} = \dfrac{BO}{OD}$ — (ii).
Equating (i) and (ii), we directly obtain $\dfrac{AO}{OC} = \dfrac{BO}{OD}$. [Hence Proved]

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Q10. Case Study 01: Shadow Triangulation of Vijay's House and City Tower

Vijay is trying to find the height of a transmission tower near his house using the properties of similar triangles. The height of Vijay's house is 20 m. At a specific time of day, Vijay's house casts a shadow 10 m long on the ground, while at the same time the transmission tower casts a shadow 50 m long, and his neighbour Ajay's house casts a shadow 20 m long. [4]

(i) **[1 Mark]** What is the height of the transmission tower?

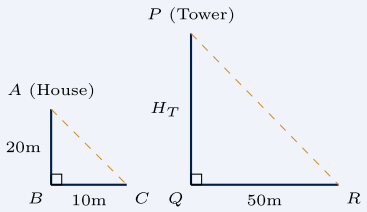
(ii) **[1 Mark]** What is the height of Ajay's house?

(iii) **[2 Marks]** When the transmission tower casts a shadow of 40 m, what will be the length of the shadow cast by Vijay's house at that instant?

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Given Data: Height of Vijay's house $AB = 20$ m; shadow of Vijay's house $BC = 10$ m; shadow of transmission tower $QR = 50$ m; shadow of Ajay's house $= 20$ m.

At any given instant, the Sun's rays strike level ground at the same angle of elevation ($\angle C = \angle R$), and all vertical structures stand perpendicular to the ground ($\angle B = \angle Q = 90°$). Thus, by the **AA similarity criterion**, all vertical object–shadow right triangles formed at the same time are similar:



Formula / Theorem Used: $\triangle ABC \sim \triangle PQR \implies \dfrac{\text{Height of Object 1}}{\text{Height of Object 2}} = \dfrac{\text{Shadow of Object 1}}{\text{Shadow of Object 2}}$

Sub-part (i): Height of the Transmission Tower [1 Mark]

Let the height of the transmission tower be $PQ = H_T$ m. Since $\triangle PQR \sim \triangle ABC$:

$$\frac{PQ}{AB} = \frac{QR}{BC} \implies \frac{H_T}{20} = \frac{50}{10} = 5 \implies H_T = 5 \times 20 = 100 \text{ m} \implies \boxed{\text{Height of Tower } = \mathbf{100} \text{ m}}$$

Sub-part (ii): Height of Ajay's House [1 Mark]

Let the height of Ajay's house be H_A m (shadow $= 20$ m at the same instant). By triangle similarity:

$$\frac{H_A}{20} = \frac{20}{10} = 2 \implies H_A = 2 \times 20 = 40 \text{ m} \implies \boxed{\text{Height of Ajay's House } = \mathbf{40} \text{ m}}$$

Sub-part (iii): Shadow of Vijay's House When Tower Casts a 40 m Shadow [2 Marks]

At the new instant, the transmission tower ($H_T = 100$ m) casts a shadow of 40 m, and Vijay's house (20 m) casts a shadow of length s m. By AA similarity of the corresponding object–shadow triangles:

$$\frac{\text{Height of tower}}{\text{Height of Vijay's house}} = \frac{\text{New shadow of tower}}{\text{New shadow of Vijay's house}} \implies \frac{100}{20} = \frac{40}{s} \implies 5 = \frac{40}{s}$$

Cross-multiplying: $5s = 40 \implies s = \dfrac{40}{5} = 8$ m $\implies \boxed{\text{Shadow of Vijay's house } = \mathbf{8} \text{ m}}$

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